👩‍🏫 Staff Guide · Biology A · Breathing, Exercise & Energy (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. Same topics and order as Ms. Dunham's class; more reading and writing than Level 2.

1. How the pieces line up

Class lesson	Level 3 piece
A5 Respiration + ATP video/ppt	Reading part 1; notes section A
Heart/Lungs; A2 Exercise lab	Reading parts 2-3; notes section B (rate math)
A3 Combustion; A4 Yeast Feast; A5 Combustion & Exercise	Reading parts 4-5; notes section C
A7 Putting it all together	Quick check + unit game

2. Supports built in

- Reading and notes each have a web copy with a 🔊 Read aloud button (highlights as it reads)
- Key-word box at the top of the notes; word problems have a worked example first
- Less filled in than Level 2: short answers, tables, a drawing
- No slide decks: students follow Ms. Dunham's slides with the Level 3 notes

3. Files

- Reading (+ 🔊 web copy)
- Guided notes + key (+ 🔊 web copy)
- Quick check + key
- Unit game: 10 questions

4. What to grade

- Guided notes stand in for the matching class notes or worksheet (same points when complete)
- Quick check: score it, then scale to the class quiz points
- Reading check questions: completion points
- Modified grading only as the IEP allows. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Class content: Ms. Dunham.

🫁 Breathing, Exercise & Energy: Guided Notes

Biology A · Unit 5 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A2 Exercise lab, A3 Combustion, A4 Yeast Feast, A5 Respiration + ATP, heart/lungs, A7

Name: _____

Use the class slides and your reading. Key words:

cellular respiration | *ATP* | *mitochondria* | *oxygen* | *carbon dioxide* | *combustion* | *yeast* | *BTB* | *heart rate*

A. Cellular respiration (A5)

1. Cellular respiration happens in the _____.

2. Fill in the table.

Inputs	Outputs
—	

3. ATP releases energy when a _____ group comes off.

B. Exercise lab (A2)

4. **Breathing rate change**

> Example: 14 breaths/min at rest, 28 after exercise → 28 − 14 = 14 more breaths per minute

a) Heart rate 70 at rest, 110 after exercise. Change = _____

b) BTB turned yellow in 90 s at rest and 45 s after exercise. How many times faster?

_____

5. What do the lab results show about exercise and CO2?

_____

_____

C. Combustion and yeast (A3, A4)

6. A burning marshmallow loses _____ because atoms leave as CO2 and water.

7. In the Yeast Feast, which tube made the most CO2 and why?

_____

_____

🫁 Breathing, Exercise & Energy: Reading

Biology A · Unit 5 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A2 Exercise lab, A3 Combustion, A4 Yeast Feast, A5 Respiration + ATP, heart/lungs, A7

Name: _____

How Cells Get Energy from Food

Your cells need energy all the time. They get it through cellular respiration: glucose + oxygen → carbon dioxide + water + ATP. ATP is the energy molecule cells can use right away, like energy coins. Energy is released when a phosphate group comes off ATP. Cellular respiration happens in the mitochondria, and both plants and animals do it.

The lungs take in oxygen and get rid of carbon dioxide. The heart pumps blood that carries oxygen and glucose to every cell. When you exercise, your muscle cells need more ATP, so breathing rate and heart rate go up.

In the exercise lab, students measured three clues of respiration at rest and after two minutes of exercise: breathing rate, heart rate and carbon dioxide. Students breathed through a straw into BTB. After exercise, the BTB turned yellow faster, showing more CO2.

Combustion (burning) is the same reaction as respiration, only fast: fuel + oxygen → carbon dioxide + water + heat and light. A burning marshmallow loses mass because its atoms go into the air as CO2 and water. Our cells do the same thing slowly and safely.

Yeast is a one-celled fungus. In the Yeast Feast lab, yeast was given glucose, starch or water. The more CO2 bubbles, the more the yeast used that food. Yeast used glucose fastest because it is already the simple sugar cells use.

Check your reading

1. Write the word equation for cellular respiration.

2. Why do heart rate and breathing go up during exercise?

3. How are combustion and respiration alike?

🫁 Breathing, Exercise & Energy: Quick Check

Biology A · Unit 5 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A2 Exercise lab, A3 Combustion, A4 Yeast Feast, A5 Respiration + ATP, heart/lungs, A7

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Cellular respiration makes…

A. ATP, CO_2 and water | B. Glucose and oxygen | C. Only heat

2. Respiration happens in the…

A. Chloroplasts | B. Stomata | C. Mitochondria

3. During exercise, heart rate…

A. Goes down | B. Stops | C. Goes up

4. BTB turns yellow when you breathe in it because of…

A. Oxygen | B. Carbon dioxide | C. Water

5. Compare combustion and cellular respiration (one way alike, one way different).

6. Where did the energy in ATP first come from?

🫁 Breathing, Exercise & Energy: Guided Notes · ANSWER KEY

Biology A · Unit 5 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A2 Exercise lab, A3 Combustion, A4 Yeast Feast, A5 Respiration + ATP, heart/lungs, A7

Name: _____

Use the class slides and your reading. Key words:

cellular respiration · *ATP* · *mitochondria* · *oxygen* · *carbon dioxide* · *combustion* · *yeast* · *BTB* · *heart rate*

A. Cellular respiration (A5)

1. Cellular respiration happens in the **mitochondria** .

2. Fill in the table.

Inputs	Outputs
glucose	**carbon dioxide**
oxygen	**water**
—	**ATP (energy)**

3. ATP releases energy when a **phosphate** group comes off.

B. Exercise lab (A2)

4. **Breathing rate change**

> Example: 14 breaths/min at rest, 28 after exercise → 28 − 14 = 14 more breaths per minute

a) Heart rate 70 at rest, 110 after exercise. Change = **40 beats per minute**

b) BTB turned yellow in 90 s at rest and 45 s after exercise. How many times faster? **2 times faster**

5. What do the lab results show about exercise and CO2?
After exercise the body makes more CO2 (BTB turned yellow faster) because cells do more respiration.

C. Combustion and yeast (A3, A4)

6. A burning marshmallow loses **mass** because atoms leave as CO2 and water.

7. In the Yeast Feast, which tube made the most CO2 and why?
Glucose: it is the simple sugar yeast can use right away.

🔑 *Key written by our team from Ms. Dunham's materials, please check.*

🫁 Breathing, Exercise & Energy: Reading · ANSWER KEY

Biology A · Unit 5 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A2 Exercise lab, A3 Combustion, A4 Yeast Feast, A5 Respiration + ATP, heart/lungs, A7

Name: _____

How Cells Get Energy from Food

Your cells need energy all the time. They get it through cellular respiration: glucose + oxygen → carbon dioxide + water + ATP. ATP is the energy molecule cells can use right away, like energy coins. Energy is released when a phosphate group comes off ATP. Cellular respiration happens in the mitochondria, and both plants and animals do it.

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Combustion (burning) is the same reaction as respiration, only fast: fuel + oxygen → carbon dioxide + water + heat and light. A burning marshmallow loses mass because its atoms go into the air as CO2 and water. Our cells do the same thing slowly and safely.

Yeast is a one-celled fungus. In the Yeast Feast lab, yeast was given glucose, starch or water. The more CO2 bubbles, the more the yeast used that food. Yeast used glucose fastest because it is already the simple sugar cells use.

Check your reading

1. Write the word equation for cellular respiration.
Glucose + oxygen → carbon dioxide + water + ATP (energy).

2. Why do heart rate and breathing go up during exercise?
Muscle cells need more ATP, so they need more oxygen and glucose and make more CO2.

3. How are combustion and respiration alike?
Both use fuel + oxygen and give off CO2, water and energy.

🫁 Breathing, Exercise & Energy: Quick Check · ANSWER KEY

Biology A · Unit 5 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A2 Exercise lab, A3 Combustion, A4 Yeast Feast, A5 Respiration + ATP, heart/lungs, A7

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Multiple choice and short answer. You may have it read aloud.

1. Cellular respiration makes…

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2. Respiration happens in the…

A. Chloroplasts | B. Stomata | **C. Mitochondria**

3. During exercise, heart rate…

A. Goes down | B. Stops | **C. Goes up**

4. BTB turns yellow when you breathe in it because of…

A. Oxygen | **B. Carbon dioxide** | C. Water

5. Compare combustion and cellular respiration (one way alike, one way different).

Alike: fuel + oxygen → CO2 + water + energy. Different: combustion is fast with heat and light; respiration is slow in cells and makes ATP.

6. Where did the energy in ATP first come from?

The sun (stored in food by photosynthesis).

🔑 *Key written by our team from Ms. Dunham's quizzes and notes, please check.*